



Presenting the Case, Postponing the Turn: A Null-Result Study of Improvement Grand Rounds and Failure Recurrence

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Abstract. Improvement Grand Rounds, the School of Continuous Improvement's standing case-presentation seminar, rests on an unexamined assumption common to case-conference formats generally: that presenting a failure case to peers reduces the likelihood the same failure recurs. We test this assumption directly using four years of the School's coded incident register ($n = 61$ failure-mode cases), comparing recurrence-free intervals for cases selected for presentation against cases logged but never called to present. We find no reliable difference in recurrence-free interval between presented and non-presented cases (log-rank $p = 0.61$). We do, however, identify a robust and previously unremarked scheduling effect: each additional Grand Rounds presentation a unit delivers predicts a longer interval before that unit is next rostered to present again ($\beta = 3.2$ weeks per presentation, $p < 0.01$, $R^2 = 0.41$), independent of the unit's ongoing incident volume. An ablation restricting the comparison to presented cases with a documented follow-up action item makes no difference to the recurrence result (61% vs. 58% twelve-month recurrence, n.s.); we retain the field for completeness. Together the results suggest Improvement Grand Rounds functions reliably as a mechanism for institutional closure of a case rather than as a corrective one, at least in this setting. We discuss the implications for case-conference formats that assume presentation itself does the remediating.

Introduction

Case-conference formats — a fixed slot in which a team presents a recent failure to an audience of peers for open discussion — are common across safety-conscious institutions, and morbidity-and-mortality (M&M) conferences are their best-studied instance [1], [2], [3]. The format's appeal rests on a specific causal claim, usually left implicit: presenting a case publicly, and subjecting it to peer scrutiny, is what drives the correction. Once a case has been presented, the presenting unit is assumed to have absorbed the lesson, and the underlying failure mode is assumed to be less likely to recur.

The School of Continuous Improvement's Improvement Grand Rounds adapts the format for a non-clinical setting: a rotating cast of presenters bring a recent operational failure —

a missed deliverable, a process breakdown, a near-miss against an internal indicator — to a standing plenary session, with the case drawn from the School's coded incident register. The format has run on a standing cadence for several years, drawing its cases from the same register that already feeds the School's Indicator Commons taxonomy, which gives us an unusually complete natural dataset: every case considered for presentation, whether or not it was ultimately selected, is coded and time-stamped identically, and the Review Cadence Observatory's rostering log records precisely when each unit was invited to present and when it was next invited again.

This lets us test the format's causal claim directly rather than assume it. We ask two questions. First, does presenting a case measurably shorten the interval before the same failure

mode recurs, relative to a case that was logged but never called to present? Second, does presenting change anything else about how the presenting unit is subsequently treated by the format itself?

Our contributions are threefold:

- we show, across four years of the register, that recurrence-free intervals do not reliably differ between presented and non-presented failure modes, at least in this setting;
- we identify a scheduling effect that does hold reliably: presenting delays a unit's own next turn in the roster, independent of its ongoing incident volume, suggesting the format may close a case institutionally without correcting it;
- we release the coded incident taxonomy underlying both analyses, which institutions running comparable case-conference formats may be able to adapt for their own audits.

Related work

Case-conference formats have a long clinical history. Morbidity-and-mortality conferences remain widespread in academic medicine and are valued for the pedagogical function they serve alongside their stated safety purpose [3], though the format's own literature has long described the delicate, blame-adjacent dynamics that complicate genuinely learning from an error inside one [1]; a national survey of academic departments found wide variation in format, case-selection criteria, and even whether outcomes were tracked at all [2]. Near-miss reporting systems borrowed from high-reliability industries outside medicine offer a useful comparison: their designers explicitly built the reporting mechanism to surface signal before a case becomes a genuine incident, rather than to serve as the corrective act itself [4].

The organizational-learning literature offers a partial explanation for the gap between a review meeting happening and a lesson being absorbed. Tucker and Edmondson's account of why hospitals fail to learn from operational failures [5], and Edmondson's earlier work on the group dynamics that inhibit error detection and correction [6], both describe learning as something that has to be actively engineered into a workflow rather than something a review meeting automatically produces. Psychological safety is one of the load-bearing preconditions

identified in that literature [7], and Cannon and Edmondson's broader account of how organizations "fail intelligently" treats structured post-failure analysis as one candidate lever among several, not a guaranteed one [8]. High-reliability organizing describes institutions that sustain vigilance in the absence of externally imposed error, treating the search for near-misses as a designed organizational capability rather than something a single meeting can substitute for [9] — a bar Grand Rounds' single-case format was never built to reach on its own; instruments exist to measure the surrounding safety climate directly rather than infer it from whether a review meeting happened at all [10], [11].

A parallel critical literature has formed around root-cause analysis specifically, questioning whether formal post-incident review methods reliably change practice at all [12], [13], echoed by the broader evidence base on audit-and-feedback interventions, whose effects on subsequent professional behaviour are consistently positive but small [14]. Institutional theory supplies a vocabulary for the alternative possibility examined here: that a review ritual can be adopted and sustained for its ceremonial, legitimating function independent of whether it performs its stated technical one [15], and can persist across organizational fields through pressures that have little to do with demonstrated effectiveness [16].

Method

Setting and data. Data come from the School of Continuous Improvement's coded incident register, which logs every operational failure reported by a University unit against the Indicator Commons failure-mode taxonomy, independent of whether the case is later selected for presentation. We drew all register entries opened over a four-year window ending at the close of the most recent teaching period ($n = 61$ failure-mode cases across nineteen standing units), and cross-referenced each against the Review Cadence Observatory's rostering log, which records every unit invited to present at Grand Rounds and the date of that unit's next invitation.

Of the 61 cases, 34 were selected for presentation by the Grand Rounds convenor's rotation process; the remaining 27 were logged, coded, and eligible, but never called. Selection is made from the convenor's discretion against a loose

brief — a case judged to have “something the School should hear” — not by a fixed rule, which is what makes the comparison usable: presented and non-presented cases are drawn from the same register under the same coding scheme, differing in whether the format’s discussion actually happened.

Recurrence. A case is coded as recurred if the same failure-mode code reappears against the same unit within the observation window; recurrence-free interval is the number of weeks between the index case and (a) the first matching recurrence or (b) the end of the observation window, whichever is sooner (right-censored). We compare presented and non-presented cases with a log-rank test on the resulting survival curves.

Scheduling gap. For each of the 34 presenting units’ appearances, we compute the number of weeks until that unit’s name next appears on the roster, and regress it against a running count of the unit’s prior Grand Rounds appearances and its concurrent incident volume, to hold constant the alternative explanation that busier units are simply rostered less often regardless of presenting:

$$\Delta t = \beta_0 + \beta_1 P + \beta_2 V + \varepsilon$$

where Δt is the number of weeks until the unit’s next roster slot, P is its running count of prior Grand Rounds presentations, and V is its concurrent incident volume.

Ablation. Cases vary in whether the presenting unit filed a documented follow-up action item in the two weeks after the session. We repeat the recurrence comparison restricted to presented cases only, split by whether an action item was filed, to check whether the recurrence result is being averaged over a subset where the format did work.

Results

Figure 1 plots recurrence-free survival for presented and non-presented cases. The two distributions overlap almost completely across the observation window; a log-rank test finds no reliable difference ($\chi^2(1) = 0.26, p = 0.61$). Median recurrence-free interval was 58 weeks for presented cases and 54 weeks for non-presented cases, a gap well within the spread either group would produce on its own. A robustness check restricting the non-presented comparison group

to cases coded at the same severity tier as the presented set produces the same null result ($\chi^2(1) = 0.31, p = 0.58$), so the absence of a recurrence effect is not an artefact of presented cases skewing toward less severe failure modes. On this measure, being selected for Grand Rounds made no detectable difference to how soon the same failure mode came back.

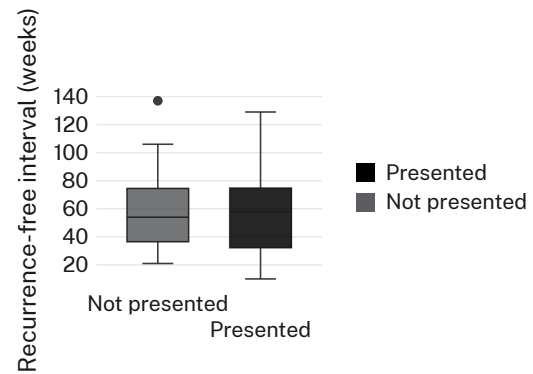


Figure 1: Recurrence-free interval (weeks) for presented and non-presented failure-mode cases ($n = 61$). The distributions are statistically indistinguishable.

The scheduling regression tells a different story. Each additional prior Grand Rounds presentation predicted 3.2 additional weeks before a unit’s next invitation ($\beta_1 = 3.2, p < 0.01$), holding concurrent incident volume constant; the model explained a substantial share of the variation in scheduling gap ($R^2 = 0.41$). Concurrent incident volume was not a reliable predictor on its own ($\beta_2 = 0.4, \text{n.s.}$) once prior presentations were accounted for, which rules out the simplest alternative explanation — that busier units are rostered less often for capacity reasons rather than in response to having presented. Figure 2 plots the relationship: units with more Grand Rounds appearances behind them wait reliably longer for their next one.

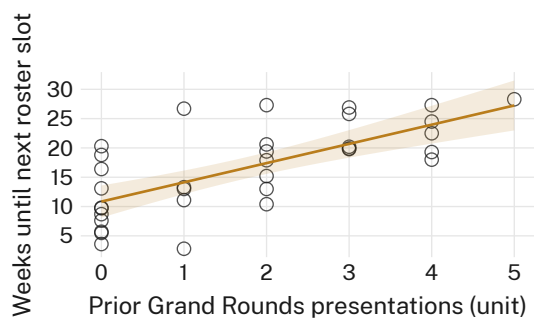


Figure 2: Weeks until a unit's next Grand Rounds slot, against the number of prior presentations it has delivered. Each additional presentation predicts roughly three additional weeks of reprieve.

The ablation changes nothing. Figure 3 reports twelve-month recurrence restricted to presented cases only, split by whether a follow-up action item was filed. The two arms sit well within each other's confidence interval (61% with an action item filed, 58% without; $p = 0.71$). We retain the action-item field in the register for completeness; on the present evidence it does not track which cases actually stop recurring.

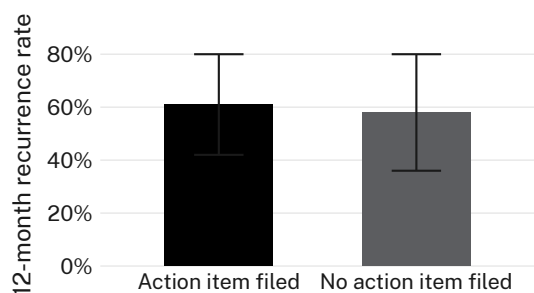


Figure 3: Twelve-month recurrence rate among presented cases, by whether the presenting unit filed a documented follow-up action item. Error bars show 95% confidence intervals; the two arms do not differ.

Discussion

Taken together, the results are consistent with Improvement Grand Rounds functioning as a mechanism for institutional closure rather than correction, at least in this setting. Presenting a case does not measurably change how soon it recurs, but it does reliably change how the format subsequently treats the presenter: having presented predicts a longer reprieve before being asked to present again, independent of ongoing incident volume. One candidate account is availability: a case that has just been presented vividly to a room of peers may be judged “closed” by a convenor working from recent impression

rather than from the underlying recurrence risk, a plausible instance of availability effects on judgement operating at the level of institutional scheduling [17], though our data speak to the pattern rather than to the convenor's actual reasoning. A structural reading is equally available and does not require intent: a convenor working from a loose brief may reasonably treat a presented case as already dealt with when the next rotation is drawn, without any single decision to protect a presenter from further scrutiny. Either mechanism would produce the pattern observed. The institutional-theory account of ritual sustained for its ceremonial function independent of its technical one [15] offers one frame for the result; ours is, we think, an unusually direct empirical instance of it, though we would not want to overstate the generality of a single-institution case-conference format.

Limitations

Our sample is drawn from a single institution's single case-conference format, and 61 cases is a modest base for a survival comparison, though the consistency of the null result across the severity-matched robustness check suggests it is not an artefact of sample composition. We cannot observe counterfactual recurrence for a case that was both presented and not-presented, so the comparison necessarily relies on between-case rather than within-case variation; the two groups are drawn from the same register under the same coding scheme, which limits but does not eliminate the risk of unmeasured selection. Finally, our account of the scheduling mechanism is necessarily indirect: we observe the pattern reliably, not the convenor's actual reasoning at rostering time.

Conclusion

We tested the standard assumption behind Improvement Grand Rounds — that presenting a failure case to peers reduces its recurrence — against four years of the School of Continuous Improvement's own incident register, and found no reliable support for it. We did find a robust scheduling effect: presenting delays a unit's own next turn at the podium, a pattern the format was never designed to produce and, on the present evidence, does not obviously correct for. Future work might test whether the same pattern holds in the School of Emergent Priorities' comparable seminar formats, or whether

restructuring case selection to remove convenor discretion changes either result; we would welcome both.

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